ENEKO RIOS ARMENTA

Physics Undergraduate — Computational Research & Data Analysis
eneko@ucsb.edu · +1(669)214-8715 · University of California, Santa Barbara
GitHub · LinkedIn · Portfolio URL

EDUCATION

University of California, Santa Barbara (UCSB) Santa Barbara, CA
B.S. in Physics, Minor in Statistics Expected June 2028
Relevant Coursework: Vector calculus, linear algebra, fundamental physics, data science principles

RESEARCH EXPERIENCE

Undergraduate Researcher — Matthieu Louis Lab, UCSB Jan 2025 – Present
- Conduct experiments investigating fruit fly larval behavioral responses to olfactory neuron stimulation.
- Process larval movement data, including body-position measurements over time, into structured datasets for quantitative analysis.
- Use Python and pandas to analyze behavioral patterns, generate visualizations, and identify observations in movement data.
- Investigate relationships between neuron stimulation and larva behavior through data-driven analysis.

PROJECTS

Exoplanet Population Analysis — Independent Project
- Analyzed publicly available data from the NASA Exoplanet Archive using Python-based scientific computing tools.
- Investigated the relationship between planetary mass and radius using statistical analysis and regression.
- Identified an empirical change in the mass–radius relationship near 10.3 Earth radii and evaluated its robustness using a subset with relative mass and radius uncertainties of 30% or less.
- Developed visualizations and considered measurement uncertainty, sample-selection effects, and limitations in interpreting the results.

TECHNICAL SKILLS

- **Programming:** Python, R
- **Data Analysis:** pandas, NumPy, SciPy
- **Visualization:** Matplotlib
- **Tools:** Jupyter Notebook, Git, GitHub, RStudio
- **Additional Technical Experience:** Hands-on automotive maintenance, including replacing headlights, sound systems, and brakes under the guidance of a mentor.

ADDITIONAL EXPERIENCE

Intramural Soccer Referee — UCSB 2025 – Present
Referee intramural soccer matches, applying rules consistently and managing game situations.

Club Soccer Referee — Los Gatos United 2022 – 2025
Refereed club soccer for three years.

Farmers Market Strawberry Stand — P and K Farms 2023 – 2024
Worked at a strawberry stand supporting day-to-day market operations and customer service.